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Study on real-time wearable sport health device based on body sensor networks

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ABSTRACT

As a branch of the wireless sensor network and an important part of the Internet of Things (IoT), Body Sensor Network (BSN) integrates biomedical engineering technology and wireless sensor network. It has high precision, small monitoring range, miniaturization and low power. The congruent feature can monitor the physiological information and motion information of the human body in real time, and provide the tester with many important quantitative information, to realize early monitoring, early prevention and early intervention of some common chronic diseases. This technology can be widely applied in biomedical, sports training, daily activity monitoring and other fields. In addition, the human sensor network is also an implementation of home health care, providing a new way of thinking for telemedicine and home health care. In order to realize the human motion monitoring system based on human body sensor network. This research aims to design a wearable sport health monitoring system with low cost, low power consumption, high modularity, high reliability, and high precision. The human body motion monitoring system of the axis wireless sensor platform enables real-time monitoring of sports health.

1. Introduction

With the improvement of people's quality of life and the improvement of the medical environment, everyone's attention to their own health is gradually strengthening, and the medical treatment method has begun to transform from a passive treatment to active prevention, health care, and pre-diagnosis [1,2]. This transformation means that the prevention and supervision of diseases with small groups as the unit will become a hot industry in the future [3,4]. However, in real practice each small group cannot be equipped with one professional doctor. Therefore, it is particularly important to provide remote information communication and technical support for non-professionals by using existing medical technology and modern technology [5]. Currently, research based on this field mainly focuses on telemedicine, personal health testing, disease files, integrated treatment process and communication between doctors and patients [6]. For example, the telemedicine home care system developed by Melisko et al. [7] reported body data tracking and statistics. Shahryar et al. [8] designed an intelligent mobile medical monitoring system using relevant sensors to obtain various physical indicators, and timely notification to patients. Jackson and other patients [9] designed a real-time monitoring for heart patients system, monitoring for specific patients. However, most of the systems only stay in the development of mobile phone clients and the transmission of information from a small number of sensors. The data acquisition process is single, and the data analysis is superficial and depthless.

Due to the importance of BSN and its far-reaching impact on future social issues such as health care and helping the elderly, more and more people are investing in BSN research [10,11]. The BSN node developed using TI's MSP430F149 microprocessor chip and CC420 RF transceiver. The software uses Berkeley's TinyOS and studies the situational awareness and ambient perception in BSN. Llosa et al. [12] used the BSN platform to study the kinematics of the thighs and waist during rowing. The research of gesture recognition is carried out. National University of Singapore and other in-depth do researches on BSN network communication. Deshpande et al. studied the asymmetric multi-hop network of gastrointestinal catheter multi-capsule communication, by simulating the energy consumption of a given transceiver circuit [13]. It is found that multi-hop networks are very effective for saving BSN node energy. Hossain et al. conducted a preliminary study on the method of eliminating inter-network interference when multiple BSN networks exist in the same area [14]. Some other researchers [15,16] focuses on the research of sensor multi-data fusion algorithms, and performs Bayesian motion reconstruction on sensor node data dynamically, which is mainly used for 3D reconstruction of human motion in animation [17,18]. Tang et al. [19] developed a real-time fall attitude and heart rate monitoring system, which conducted an alarm test on the normal fall phenomenon and obtained a correct rate of more than 90% of the fall alarm.

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In order to adapt to this change of health monitoring form, a personalized health supervision platform is designed. This paper combines IoT technology, information technology, data processing and analysis technology to design a set of universal self-service information collection equipment that can be used openly. The terminal constructs a human motion monitoring system based on a nine-axis wireless sensor platform, including hardware, embedded software and host computer software. The common data filtering method of motion sensor is analyzed and compared with the walking acceleration signal: median filtering, low-pass filtering, wavelet packet transform and Kalman filtering. Kalman filter is a data processing algorithm that optimizes autoregression. It is mainly used to solve the estimation problem in linear systems. The essence is to find the estimated value under the minimum mean square error. Its characteristic is to use the recursive method to calculate the actual value, and the estimation process data storage is relatively small, which is convenient for the system to update the data quickly and in real time. In general, the Kalman filter uses the state prediction equations and observation equations to accurately estimate the state of the system's data in real time. The preliminary results show that the Kalman filter has good performance in real-time, delay and denoising performance [20,21]. The Kalman filter is most suitable for the hardware platform under low-speed motion. In the attempt to walk, the double pendulum model was used to obtain the walking displacement by using the change of the component of the lower limb in the forward direction at the adjacent time, and a certain effect was obtained. The rest of this paper is organized as follows. Section 2 discusses the overall design of health monitoring system including the communication protocol, followed by the hardware and software design in Section 3. The results of system noise and posture error are discussed in Section 4. Section 5 concludes the paper with summary and future research directions.

2. Overall design of health monitoring system

2.1. Overall frame design

The design of the motion monitoring system based on human body sensor network is designed with low cost, low power consumption, high modularity, high reliability, high precision and easy wearing. The motion monitoring system of this paper consists of hardware platform, monitoring software and communication protocol. The overall structure of the system is shown in Fig. 1. The hardware system includes: a sensor node, a communication base station, a battery and an antenna, and the core hardware is a sensor node and a communication base station. Each sensor node is responsible for the acquisition, storage and transmission of motion data for the respective parts. A plurality of sensor nodes can be set as a central node, which is responsible for the work of sensor node networking and control. The communication base station is responsible for the start and end of the motion data acquisition, and receives the data of the sensor node and uploads it to the personal computer for data processing and storage. The monitoring software is divided into client software and server software. The client software can be used for home monitoring separately, and the server software can realize remote monitoring. The functions implemented by the client software and the server software are basically the same, and the interfaces implemented by the two are basically the same. The main differences are:

• Client software displays data in real time from the sensor node, while the server software displays data in real time from the data uploaded by the client to the server

• Server software provides enhanced data management capabilities.

The client software adopts MFC multi-document multi-view architecture, which can dynamically display the data of each sensor node, provide communication configuration and data analysis functions of base station nodes, and perform real-time filtering processing. Real-time calculation of attitude angle for 3D visual display and driving human bones Exercise and walking; provide uploading filter data to the server, multi-node offline data playback function; provide simple database operation, perform data archiving, query, data visualization and other functions; provide functions such as printing raw data and analysis results. The server software also uses MFC multi-document multi-view architecture to dynamically display the filter data uploaded by the client. In order to calculate the attitude angle in real time, three-dimensional visual display and drive human bone movement and walking is performed which provides multi-node offline data playback function, powerful database operation, perform data archiving, query, delete, data visualization and other functions. The communication protocol includes data transmission and reception communication between the node and the base station, synchronization of multi-node data collection, communication mode between nodes, and connection topology.

2.2. Communication protocol design

The communication part of the system mainly consists of three parts: node and base station, node-to-node communication, base station-to-client communication, and client–server communication. The communication between the node and the base station belongs to the wireless network communication. The communication between the base station and the client belongs to the serial port communication, and the communication between the client and the server belongs to the wired network communication [22,23]. There are three common BSN network topologies: point-to-point communication, single-hop star network and network multi-hop structure. The advantage of the point-to-point communication method is that the network complexity is small, the protocol is sold less, and the number of layers of the network is less real-time. The disadvantage of this communication method is that the distance between the node and the base station is limited, it is impossible to be too far, and the radiation to the human body is also larger [24]. In a single-hop star network, each object has a central node responsible for data relay from the node to the base station and control from the base station to the node. The advantage of this communication method is that not all nodes need a large transmission power, and the harm to the human body is small, and the transmission distance is farther than the point-to-point method [25,26]. The network type multi-hop structure uses the human body as the transmission medium, and the data of each node jumps from the neighboring node to the central node through the optimal routing path. Such a network has lower power requirements for the node, but the routing complexity is high. In this paper, the wireless communication protocol design has not been studied in depth, and the simplest point-to-point method is adopted.

The communication between the base station and the client adopts a USB/RS232 bridging method. If USB communication is used, the development of the underlying USB driver is costly and difficult; if the conventional RS232 is used, the base station is larger and heavier, and an additional RS232 cable is required. For the sake of comprehensive consideration, the USB/RS232 bridge method is used. The client transmits the communication command to the base station through the serial port, and then the base station wirelessly transmits the signal to the node. Table 1 shows the communication commands. The client needs to upload the data to the server for storage, and the data dynamically displayed by the server is also from the client.

3. Design of hardware and software systems

3.1. Hardware system

The sensor nodes in this paper are mainly composed of microcontroller, motion sensor, wireless module, power management module, flash and so on. Since the node needs to be small and must be able to work continuously for a long time, the microcontroller must support the

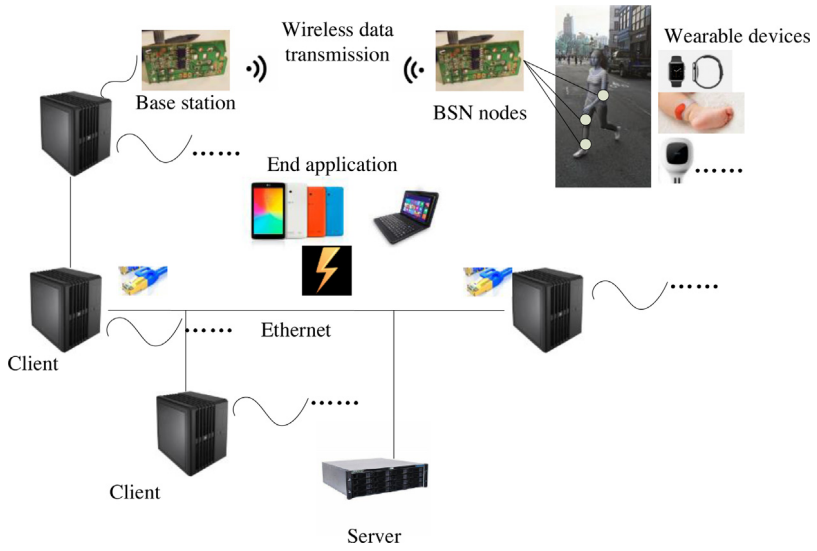


Fig. 1. Schematic of overall system design.

Table 1
Communication command.

Command	Explanation
02 00 XX XX XX XX 03	Real-time data uploading mode, xx is data
02 01 XX XX XX XX 03	Offline data uploading mode, xx is offline package No.
02 02 XX XX XX XX 03	Change nRF905 node address, xx is target address
02 03 XX XX XX XX 03	Change nRF905 node frequency, xx is target frequency
02 04 XX XX XX XX 03	Erase Flash, xx can be any data
02 05 XX XX XX XX 03	Change nRF905 host address, xx is target address
02 06 XX XX XX XX 03	Change nRF905 host frequency, xx is target frequency

low-power mode. The node integrates a nine-degree-of-freedom motion sensor, and the amount of data that needs to be processed is large. In order to obtain information on human motion, the system requires a motion sensor. With the development of MEMS technology, motion sensors can be made very small and with high precision. The motion sensor of the node has a total of nine degrees of freedom, namely a three-axis accelerometer, a three-axis gyroscope and a three-axis magnetometer. The three-axis accelerometer outputs the acceleration values of the X, Y, and Z axes, the three-axis gyroscope outputs angular velocity values around the X, Y, and Z axes, and the three-axis magnetometer outputs the magnetic field strengths of the X, Y, and Z axes. These nine parameters can be combined to determine the activity of the wearing part of the node. The main object of the hardware design of the platform is the intelligent wearable device, which mainly realizes the information collection of various data of the body. Each module in the smart wearable device is equipped with a power supply, which can be disassembled or assembled by the user. These health data can be selectively transmitted to the terminal application or the medical information processing server, when the selection module is used independently. The sensor data will be automatically uploaded to the terminal application through the BSN. Otherwise, the sensor will upload the data to the medical information processing server according to the current environment. Fig. 2 is the design framework of the wearable device. The sensors and communication elements are incorporated in the circuit. Electronic component layout tools design the layout of them.

The specific design of the algorithm is as follows:

(1) Assuming that the waveform signal is acquired at a frequency of 1 ms, the peak value M_p and the trough value M_t of its signal are updated each time. Assuming that the threshold is M_{thread} and the heart beat is M_b:

$$M_{thread} = (M_p - M_t)/2 + M_b \tag{1}$$

(2) When the voltage value of the waveform signal is between the threshold range and the peak M_p, and the duration of the heartbeat is more than 60%, the signal can be set to the heartbeat value M_h, and at the same time the two timers T_1 and T_2 are turned on. It is based on the time difference between the 2nd and 1st heartbeat and the difference of the T_1 time (the effective heartbeat interval). When no new heartbeat value occurs within the specified time, the waveform information is restored to the initial value and recalculated.

(3) According to step (2), the mean $M_{average}$ of 20 consecutive heart beats is counted. If the number of heart beats is N_i during this period:

$$H_1 = [N_1, N_2, N_3, \dots, N_i] \tag{2}$$

(4) During the total time T, the voltage value of the pulse signal exceeds M_{thread} the number N_i. The equation is achieved:

$$H_1 = [M_1, M_2, \dots, M_i] \bullet 60/T_{total} \tag{3}$$

(5) Using Eq. (2) to perform median filtering, remove the singularity and compare it with the elements in the H_2 array to remove the less accurate data in H_2 and calculate the mean of the H_1 array, the average heart rate P mean is obtained. Each element in H_1 is compared with P mean to obtain a ΔP array:

$$P = P_{mean} + \Delta P \tag{4}$$

Table 2 shows the experimental comparison data among the heart rate calculation method, the traditional heart rate calculation method and the real values. In traditional method, the heart beat threshold is constant and the algorithm does not use median filtering. It can be seen that in the 10 groups of statistics, the difference is controlled within 6% relative to the values using traditional method. The values using proposed one is much closer to the real values. The corresponding difference is small, and the accuracy is better.

3.2. Software system

The main functions of the system node are:

(1) Receiving the base station command and parsing, and making corresponding responses.

(2) Performing flash programming in real time and backing up the sensor data.

Therefore, the corresponding programming modules involved in the node include MCU initialization, motion sensor, ADC sampling, timer, SPI communication, nRF905 transceiving data, and flash memory. Due

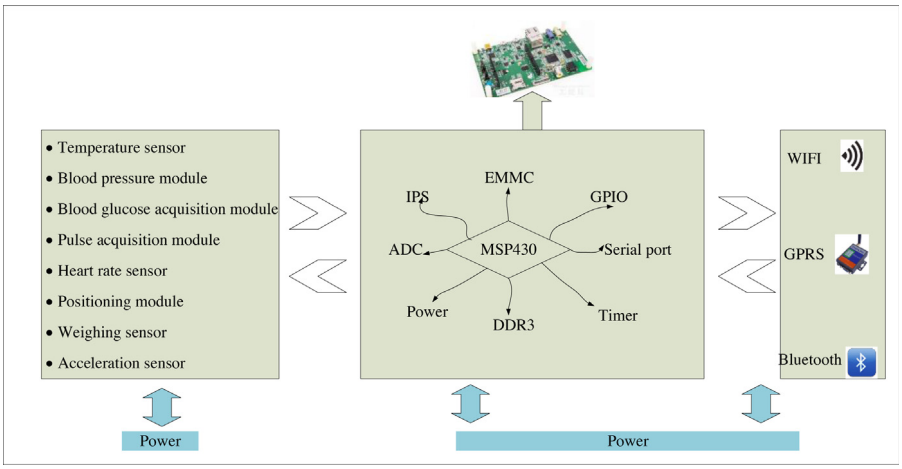


Fig. 2. Schematic of hardware system design.

Table 2
Comparison of heart rate calculations.

Real value (counts/time)	Traditional method (counts/time)	Proposed method (counts/time)	Difference to real value (%)	Difference to value using traditional method (%)
59	62	60	1.69	−3.33
65	68	65	0.00	−4.40
75	72	75	0.00	4.17
70	75	71	1.43	−5.30
69	66	68	−1.45	3.03
81	84	81	0.00	−3.57
72	76	72	0.00	−5.26
67	70	68	1.75	−2.86
82	80	83	1.22	3.75
67	65	68	1.49	4.62

to the instability of wireless communication, data packet loss is inevitable. Therefore, the real-time collection needs to upload the lost data packet from flash. The speed of offline uploads affects the time required for motion data collection. If there are many lost packets and the offline upload speed is slow, the time of one data collection will become very long, which will affect the progress of subsequent analysis. Reducing the packet loss rate can be improved by hardware (improving the antenna design) or software (using frequency hopping, which will be described later). If searching orderly, $N \bullet M$ times at most is needed. In the case of equal probability, you need to compare $(1 + N \bullet M)/2$ in average. If the last page or bytes of each page are stored in the Hash backup data to store a specific page identifier, you can perform a binary search. The half-finger search needs $\log_2(N \bullet M)$ times, and the average number of comparison times is O ($\log_2 N$), which greatly reduces the number of investigations and improves the offline upload speed. The main functions of the system base station are:

(1) Communicating with a personal computer (accepting a host computer command, uploading a data packet to a personal computer, etc.).

(2) Communicating with a node (sending a communication command, accepting node data, frequency hopping control, etc.).

The pseudo code is as follows:

If receive a command from the host computer
Then upload data package
    If node is found
    Then send information
Else wait

In addition to normal data transmission and transmission, and configuration of base station nodes, a frequency-hopping module is added in the main program to solve the problem of excessive packet loss rate.

4. Results and discussion

4.1. System noise analysis of the sensor

Due to the complexity and variability of the motion monitoring environment, the output signal of the motion sensor is often subject to various noise interferences, resulting in an inaccurate attitude angle. For this reason, the noise interference of the signal must be considered before the attitude angle calculation. Observing the original signal of acceleration and magnetometer, it is found that the noise interference of the acceleration signal is the largest. Common motion sensor signal filtering methods include median filtering, low-pass filtering, wavelet packet transform, and Kalman filtering. Mathie et al. [27] uses a median filtering method to filter the acceleration signal to identify the human motion category. It is mentioned that the human body motion frequency is concentrated at 0–20 Hz, and Huang et al. used low-pass filtering method to filter out acceleration noise above 20 Hz. In order to perform gait recognition, an 8-scale Daubechies wavelet packet transform is used for acceleration filtering. Sabatini [28] used Kalman filtering to filter out the acceleration noise of the human body interactive sports equipment. This paper designs experiments to compare the filtering effects of the above four methods on this platform. Since it is difficult to obtain real human motion signals, it is very difficult to accurately and quantitatively evaluate the four filtering methods. Here, the calibration acceleration signal of the MTx output is used as the gold standard signal. The MTx is a compact and accurate three-degree-of-freedom attitude tracker that provides drift-free 3D attitude angle and motion data: triaxial acceleration, magnetometer and gyroscope signals. The MTx's three-axis acceleration module has a low noise density of 0.002 m/s^2 WHz and is capable of outputting a calibrated triaxial acceleration signal. Twelve volunteers (8 boys and 4 girls, with an average age of 26 ± 1.7 years) were recruited for normal human walking experiments. Before the experiment starts, set the sampling rate of MTx to 50 Hz, which is the same as the sampling rate of the nine-axis motion sensor node; then align the MTx with the node coordinate axes, tightly bundle them together, and place them on the horizontal desktop, respectively. With the Y and Z axes facing up, record the corresponding data. Analyze the data. If the mean deviation of the corresponding coordinate acceleration signals is less than 20 mg, the coordinate axes of the two axes are considered to be aligned. Otherwise, the relative positions of the two are continuously adjusted until the condition is met. After that, put the bundled equipment with the sports tape about 20 cm above the tester's right disc joint, record a piece of data, check whether the two axes are still aligned, if they are no longer aligned, continue to adjust the two axes according to the above criteria. The relative position until the alignment, if still aligned, first let the

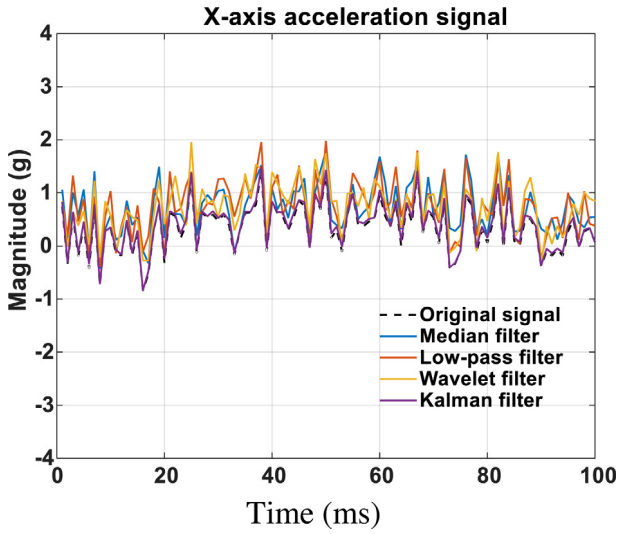


Fig. 3. Filter for X-axis signal (original signal is in black dotted line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

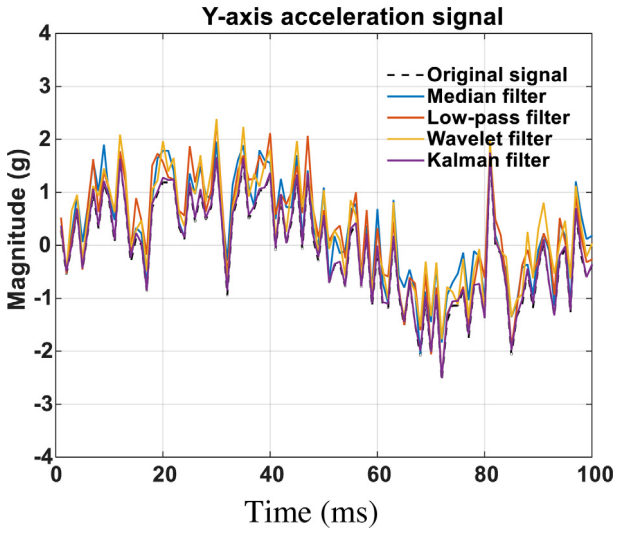


Fig. 4. Filter for Y-axis signal (original signal is in black dotted line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

tester stand for 5 s, then walk 3 times, about 10 s each time. In this experiment, the signal-to-noise ratio (SNR) and correlation coefficient (R) are selected to evaluate the filtering effects of the four filtering methods. The definition of SNR is as follows:

$$SNR = 10 \bullet \log \frac{\sum_{n=0}^{N-1} |y_n|^2}{\sum_{n=0}^{N-1} |x_n - y_n|^2} \tag{5}$$

The definition of correlation coefficient is

$$R = \frac{\sum_{n=0}^{N-1} (x_n - \tilde{x})(y_n - \tilde{y})}{[\sum_{n=0}^{N-1} (x_n - \tilde{x})^2 \sum_{n=0}^{N-1} (y_n - \tilde{y})^2]^{0.5}} \tag{6}$$

where x_n and y_n are the signal after filtering and reference signal, x and y are the average signal.

Figs. 3, 4 and 5 show the filtered image of the original acceleration signal for the four filtering methods. The red part represents the filtered signal of the node acceleration original signal, and the blue represents the calibrated acceleration signal of the MTx output. The correlation coefficients R between the node X, Y and Z axis original signals and the corresponding MTx output signals are 0.94 ± 0.02, 0.83 ± 0.03 and 0.73 ± 0.09, respectively. The signal-to-noise ratio (SNR) and correlation coefficient (R) is to facilitate a comprehensive understanding of the filtering effects of the four filtering methods.

Experimental results as shown in Fig. 6 show that Kalman filtering has the largest SNR and R values. The SNR and R values of the wavelet packet transform and the median filter are very close. Butterworth low-pass filtering has the lowest SNR and R values, and the Z-axis SNR value even has a negative value. This anomaly may be caused by a delay in the low-pass filtered waveform. Delays can also be observed. If this delay is artificially corrected, the SNR values of the low-pass filter are 44.72 ± 1.17, 27.51 ± 0.91, and 12.13 ± 0.71, respectively, and the R values are 0.99 ± 0.02, 0.97 ± 0.84 ± 0.05, respectively. The result of low-pass filtering correction delay is even better than the result of Kalman filtering. Kalman filtering only predicts the current state by means of previous data, which has better real-time performance and smaller waveform delay. The median filter has a small waveform delay, which can be observed from Figs. 3∼5, but its real-time performance is related to the selected window size N because it must wait until N/2 data points. Wavelet packet transform has a small waveform delay, but it must wait for $2''$ future data points, J is the decomposition level, and the upper computer implementation of wavelet packet transform is more difficult than the other three filtering

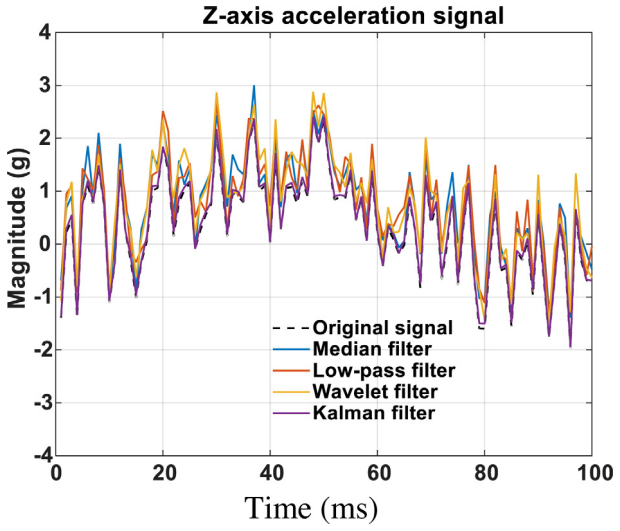


Fig. 5. Filter for Z-axis signal (original signal is in black dotted line). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

methods. As for Butterworth low-pass filtering, it is a causal system, and does not need to wait for future data points, but if the filter order and cutoff frequency are not reasonable, it will bring a large waveform delay. In this experiment, when the filter order is 3, when the cutoff frequency is 4 Hz, the X, Y and Z axis signal waveform delay is about 80 ms (4 data points). The Kalman filter is obtained experimentally and is the conclusion of the optimal filtering method for low speed motion of the platform. This method is implemented on the host computer to provide a more accurate sensor signal for attitude angle calculation. Kalman filter technology is a set of stochastic system state optimal linear filter algorithm given in recursive relationship, in target tracking, stochastic control, process monitoring. It has a wide range of applications in many fields such as fault diagnosis. In the basic Kalman filter equation, it is required that the system noise and the measurement noise are mutually uncorrelated zero-average noise. However in actual cases, system noise and measurement noise are often not white noise, but often color noise sequence. There are three cases of colored noise Kalman filter:

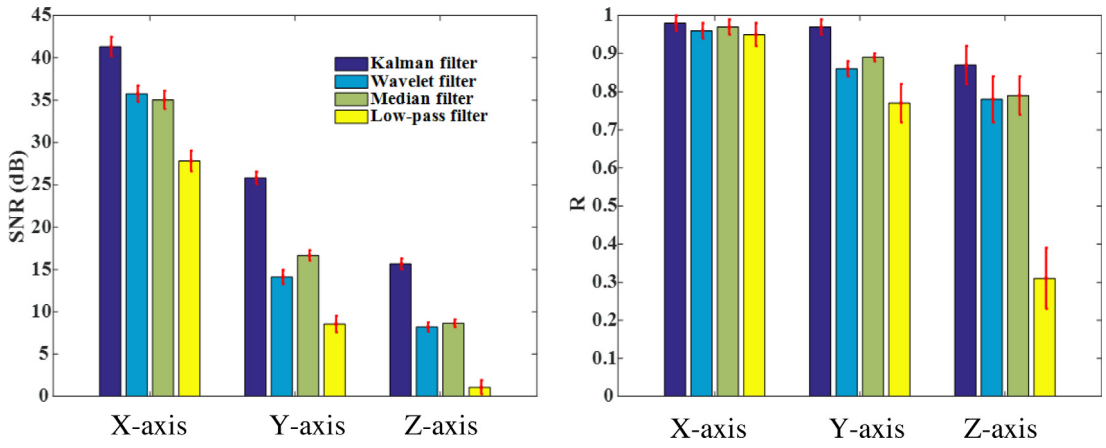


Fig. 6. SNR (left) and R (right) values with standard deviation under four filter methods.

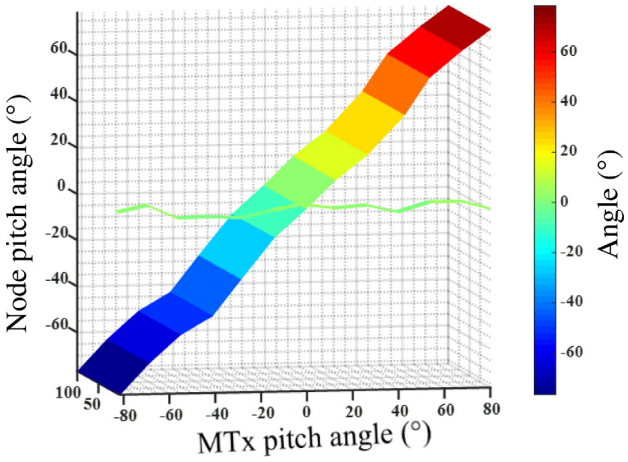


Fig. 7. Error analysis of pitch angle (thin line denotes error).

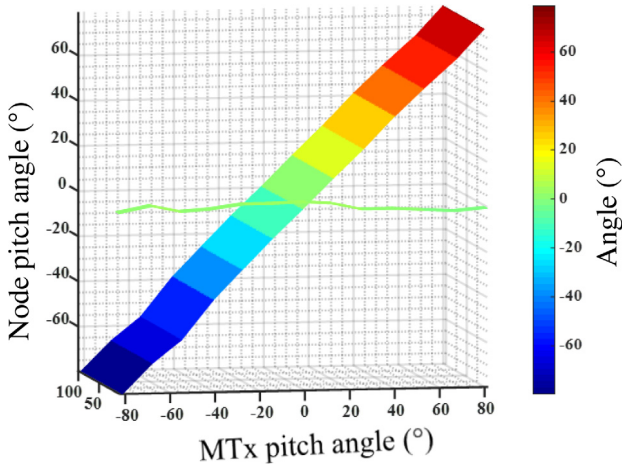


Fig. 8. Error analysis of roll angle (thin line denotes error).

(1) System noise is colored noise and observation noise is white

(2) System noise is white noise, the observation noise is colored noise

(3) System noise and the observed noise are both colored noise.

In this paper, the Kalman filter with system noise as colored noise and observation noise as white noise is applied with better results.

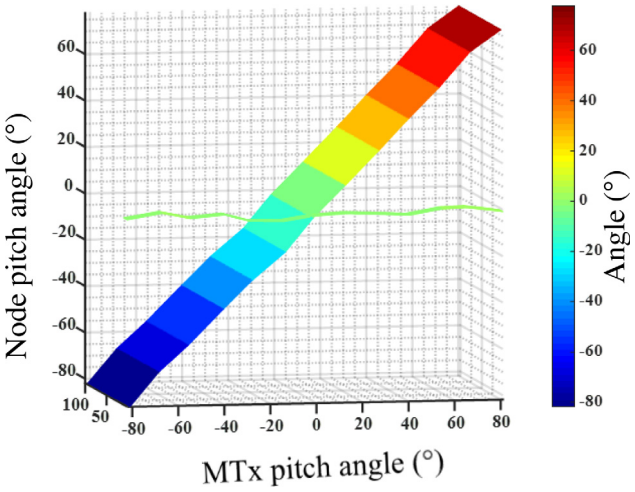


Fig. 9. Error analysis of course angle (thin line denotes error).

4.2. Attitude sensor error analysis

The carrier obtains the attitude angle by the combination of the accelerometer and the magnetometer during stationary or low-speed motion. The acceleration signal can be directly used to calculate the elevation angle and the roll angle after being filtered by the Kalman filter. If the carrier is in a non-low-speed motion state, the accelerometer and the magnetometer can still be used to obtain the attitude angle. As long as the low-pass filter extracts the gravity component, this will cause a large delay. For this reason, when the carrier continuously moves rapidly, the gyro signal is used to obtain the attitude angle.

(1) Static/low-speed error

The nodes are aligned with the MTx coordinate axes and slowly rotated around the X, Y, and Z axes to a predetermined set of reference angles (MTx output angles), and the attitude angle data output by the nodes is observed and recorded. The combination of accelerometer and magnetometer is used to find the attitude angle. Figs. 7, 8 and 9 show the error analysis of the node pitch angle, roll angle and heading angle, respectively. The red curve is the error curve and the blue curve is the corresponding attitude angle. Since the output range of the node heading angle is 0–360° and the MTx heading angle output range is −180–180°, the node heading angle is processed to make the heading angle before drawing the heading angle error analysis diagram. The range is the same as MTx. The average errors of pitch angle, roll angle and heading angle are calculated to be 2.5°, 3.3° and 3.7°, respectively;

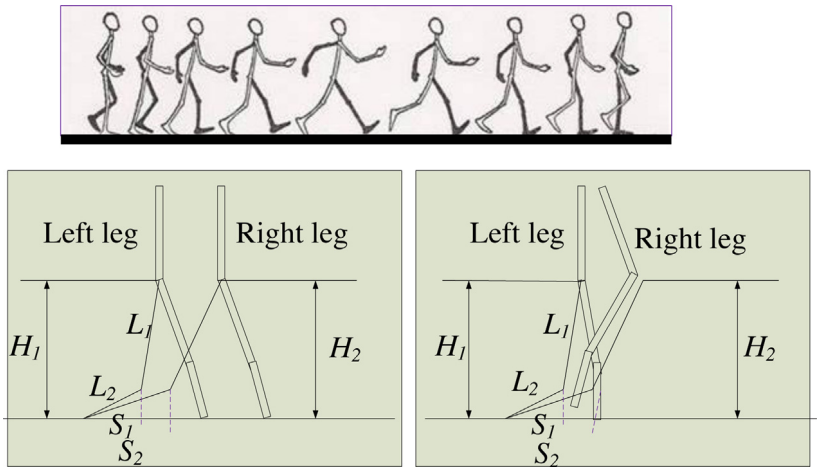


Fig. 10. The double pendulum clock model to estimate the displacement.

the maximum errors of pitch angle, roll angle and heading angle are 5.1°, 6.3° and 7.2°, respectively.

(2) Dynamic error

The nodes are aligned with the MTx coordinate axes and rotated continuously around the X, Y and Z axes respectively. The attitude estimation is mainly based on the gyroscope integration. Experiments have found that as the length of continuous rotation increases, the pose estimation becomes less accurate. Similarly, the node heading angle is processed to a range of −180–180° before the heading angle is drawn. The average errors of pitch angle, roll angle and heading angle are calculated to be 3.8°, 4.8° and 5.6°, respectively; the maximum errors of pitch angle, roll angle and heading angle are 9.1°, 10.9° and 12.1°, respectively.

The tester mainly performs low-speed leg stretching, bending legs and the like, and the attitude angle is obtained by the combination of an accelerometer and a magnetometer. The tester stands on the left leg, and the right leg extends forward at a certain angle. The human skeleton of the liquid crystal display is also standing on the left leg and the right leg is extended forward by a corresponding angle, which better reflects the movement of the lower limbs, and the delay less than 0.5 s is also better. Fig. 10 shows the process: the left and right legs are just beginning to touch the ground and perpendicular to the ground until the left foot is at the maximum angle. In this process, using the left foot to estimate the distance is not appropriate. This is because the left leg has a process of bending backwards to straight forward during the forward stroke. However, it is more appropriate to use the right foot to estimate the displacement of the movement, because the displacement of the right leg is just due to the forward movement of the center of gravity, and the right lower leg and the right thigh are always moving forward.

5. Conclusion

In this paper, aiming at realizing the wearable motion monitoring system based on Internet of Things, the hardware design and embedded software design of the nine-axis wireless motion sensor platform and the upper computer program design of the motion monitoring system are introduced in detail. The acceleration signal is taken as an example for discussion. The filtering method of motion sensor data is used to calculate the real-time attitude angle by using acceleration, magnetometer combination and gyroscope signal integration. Finally, the real-time attitude angle information is used to make a simple attempt on human bone movement and walking. The purpose of real-time monitoring of human motion is achieved. The nine-axis wireless motion sensor platform is the hardware foundation of the human motion monitoring system. The platform is divided into two parts: node and base station, including three-axis acceleration, three-axis magnetometer, three-axis gyroscope, nRF905 wireless transceiver, MSP430F2418 microcontroller, etc. The module provides continuous nine-degree-of-freedom motion data in real time. In this paper, the normal human walking experiment is designed. The commonly used data filtering methods of motion sensors are analyzed and compared with the measured walking acceleration signals: median filtering, low-pass filtering, wavelet packet transform and Kalman filtering. The following preliminary conclusions are drawn:

(1) Kalman filter has good performance in real-time, delay and denoising.

(2) Kalman filter is most suitable for the hardware platform under low-speed motion.

In this paper, two methods of acceleration, magnetometer combination and gyroscope signal integration are used to obtain the real-time attitude angle. Two simple experiments are designed to analyze the error of the two methods, and the real-time attitude angle calculation method is used to perform the human skeleton. The attempt to exercise and walk has initially achieved the goal of real-time motion monitoring.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Jiayi Zhao: Data curation. **Guangxue Li:** Formal analysis.

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